

1. Question

Develop a “Student Utility Program” that performs the following operations using a menu-driven approach.

Part A: Java Basics

Display student information.

Accept Name, Roll Number, and Marks.

Calculate Total and Percentage.

Display the result.

Part B: Conditional Statements

Implement menu options to:

Check whether a number is Even or Odd.

Find the Largest of Three Numbers.

Display Grade based on Percentage using if-else.

Display Day of the Week using switch.

Part C: Looping Statements

Write programs to:

Print multiplication tables.

Display numbers from 1 to N.

Calculate the sum of first N natural numbers.

Generate the Fibonacci series.

Part D: Methods

Create user-defined methods for:

Finding the factorial of a number.

Checking whether a number is prime.

Finding the maximum of two numbers.

Calculating the area of a circle.

Requirement: Every method should be parameterized and should return the result. Each method should be called from the main() method.

2. Steps for Execution

1. Install Java JDK on the computer.
2. Open a Java IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code.
3. Create a Java file named:
4. `StudentUtilityProgram.java`
5. Enter the Student Utility Program.
6. Save the program.

Compile the program:

```
javac StudentUtilityProgram.java
```

7. Run the program:

```
java StudentUtilityProgram
```

8. The main menu is displayed with four parts:

Part 1 – Java Basics

Part 2 – Conditional Statements

Part 3 – Looping Statements

Part 4 – Methods

9. Enter the required part number.
10. Select the required operation from the corresponding submenu.
11. Enter the required input values.
12. The selected operation is performed.
13. The result is displayed on the console.

3. Algorithm

Step 1

Start the program.

Step 2

Create a Scanner object to accept input from the user.

Step 3

Display the main menu:

1. Part 1 – Java Basics
2. Part 2 – Conditional Statements
3. Part 3 – Looping Statements
4. Part 4 – Methods

Step 4

Read the user's choice.

Step 5 – Part A: Java Basics

If the user selects Part 1:

Display student information.

Accept student name, roll number and marks.

Calculate total:

$$\text{Total} = \text{Marks1} + \text{Marks2} + \text{Marks3}$$

Calculate percentage:

$$\text{Percentage} = (\text{Total} / 300) \times 100$$

Display the student result.

Step 6 – Part B: Conditional Statements

If the user selects Part 2:

Check whether a number is even or odd using %.

Find the largest of three numbers using if-else.

Determine the grade based on percentage.

Display the day using switch.

Step 7 – Part C: Looping Statements

If the user selects Part 3:

Read a number and generate its multiplication table using for.

Read N and display numbers from 1 to N.

Calculate the sum of the first N natural numbers.

Generate the Fibonacci series.

Step 8 – Part D: Methods

If the user selects Part 4:

Find the factorial of a number.

Check whether a number is prime.

Find the maximum of two numbers.

Calculate the area of a circle.

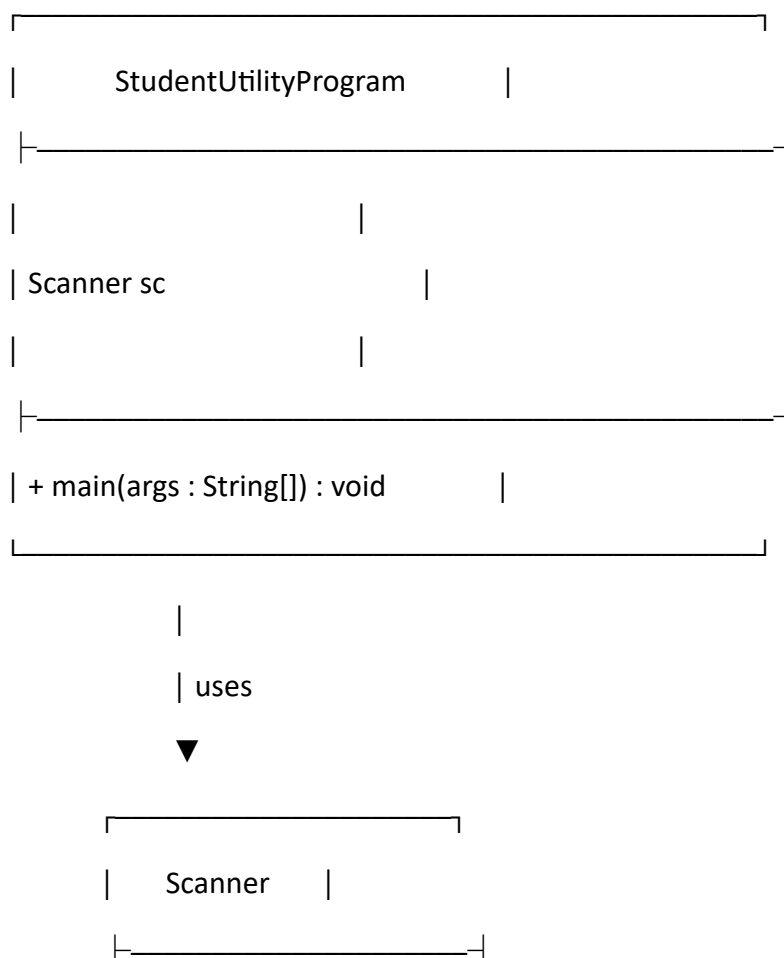
Step 9

Display the result.

Step 10

Stop the program.

UML CLASS DIAGRAM:



```
| + nextInt()      |  
| + nextDouble()   |  
| + nextLine()     |  
|_____|
```

CODE :

```
import java.util.Scanner;  
  
public class StudentUtilityProgram {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("1. Part 1");  
        System.out.println("2. Part 2");  
        System.out.println("3. Part 3");  
        System.out.println("4. Part 4");  
        System.out.print("Enter your choice: ");  
        int choice = sc.nextInt();  
        switch (choice) {  
            case 1:  
                System.out.print("Enter case 1-4 ");  
                int achoice = sc.nextInt();  
                String name="";  
                int rollno;double marks1,marks2,marks3,total,percentage;  
                switch (achoice) {  
  
                    case 1:  
                        System.out.println("Student Information");
```

```
System.out.println("Name : Sameera");
```

```
System.out.println("Roll No : 102");
```

```
break;
```

case 2:

```
System.out.println("Enter Student Name: ");
```

```
name = sc.nextLine();
```

```
System.out.println("Enter Roll Number: ");
```

```
rollno = sc.nextInt();
```

```
System.out.println("Enter English Marks: ");
```

```
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");
```

```
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");
```

```
marks3 = sc.nextDouble();
```

```
break;
```

case 3:

```
System.out.print("Enter English Marks: ");
```

```
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");
```

```
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");
```

```
marks3 = sc.nextDouble();
```

```
total = marks1 + marks2 + marks3;
```

```
percentage = (total / 300) * 100;
```

```
System.out.println("Total = " + total);
```

```
System.out.println("Percentage = " + percentage);
```

```
break;
```

case 4:

```
System.out.print("Enter Student Name: ");
```

```
name = sc.nextLine();
```

```
System.out.print("Enter Roll Number: ");
```

```
rollno = sc.nextInt();
```

```
System.out.print("Enter English Marks: ");
```

```
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");
```

```
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");
```

```
marks3 = sc.nextDouble();
```

```
total = marks1 + marks2 + marks3;
```

```
percentage = (total / 300) * 100;
```

```
System.out.println("\n----- RESULT -----");
```

```
System.out.println("Name : " + name);
```

```
System.out.println("Roll No : " + rollno);
```

```
System.out.println("Total : " + total);
```

```
System.out.println("Percentage : " + percentage);
```

```
break;
```

```
default:
```

```
System.out.println("Invalid Choice");
```

```
}
```

```
break;
```

```
case 2:
```

```
System.out.println("enter case 1-4");
```

```
int bchoice = sc.nextInt();
```

```
switch (bchoice) {
```

```
case 1:
```

```
System.out.print("Enter a number: ");
```

```
int n = sc.nextInt();
```

```
if (n % 2 == 0)
    System.out.println("Even Number");
else
    System.out.println("Odd Number");
break;
```

case 2:

```
System.out.print("Enter first number: ");
int a = sc.nextInt();

System.out.print("Enter second number: ");
int b = sc.nextInt();

System.out.print("Enter third number: ");
int c = sc.nextInt();

if (a >= b && a >= c)
    System.out.println("Largest = " + a);
else if (b >= a && b >= c)
    System.out.println("Largest = " + b);
else
    System.out.println("Largest = " + c);
break;
```

case 3:

```
System.out.print("Enter percentage: ");
```

```
double per = sc.nextDouble();
```

```
if (per >= 90)
```

```
    System.out.println("Grade A");
```

```
else if (per >= 75)
```

```
    System.out.println("Grade B");
```

```
else if (per >= 60)
```

```
    System.out.println("Grade C");
```

```
else if (per >= 40)
```

```
    System.out.println("Grade D");
```

```
else
```

```
    System.out.println("Fail");
```

```
break;
```

case 4:

```
System.out.print("Enter day number (1-7): ");
```

```
int day = sc.nextInt();
```

```
switch (day) {
```

```
    case 1:
```

```
        System.out.println("Monday");
```

```
        break;
```

```
    case 2:
```

```
System.out.println("Tuesday");
```

```
break;
```

```
case 3:
```

```
System.out.println("Wednesday");
```

```
break;
```

```
case 4:
```

```
System.out.println("Thursday");
```

```
break;
```

```
case 5:
```

```
System.out.println("Friday");
```

```
break;
```

```
case 6:
```

```
System.out.println("Saturday");
```

```
break;
```

```
case 7:
```

```
System.out.println("Sunday");
```

```
break;
```

```
default:
```

```
System.out.println("Invalid Day");
```

```
}
```

```
break;
```

```
}
```

```
case 3:
```

```
System.out.println("enter case 1-4");
```

```
int cchoice = sc.nextInt();
```

```
switch (cchoice) {
```

```
    case 1:
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        System.out.println("Multiplication Table:");
```

```
        for (int i = 1; i <= 10; i++) {
```

```
            System.out.println(num + " x " + i + " = " + (num * i));
```

```
        }
```

```
        break;
```

```
    case 2:
```

```
        System.out.print("Enter N: ");
```

```
        int n = sc.nextInt();
```

```
        System.out.println("Numbers from 1 to " + n + ":");
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print(i + " ");
```

```
        }
```

```
        break;
```

```
    case 3:
```

```
System.out.print("Enter N: ");
```

```
int m = sc.nextInt();
```

```
int sum = 0;
```

```
for (int i = 1; i <= m; i++) {
```

```
    sum = sum + i;
```

```
}
```

```
System.out.println("Sum = " + sum);
```

```
break;
```

case 4:

```
System.out.print("Enter number of terms: ");
```

```
int terms = sc.nextInt();
```

```
int a = 0, b = 1, c;
```

```
System.out.println("Fibonacci Series:");
```

```
for (int i = 1; i <= terms; i++) {
```

```
    System.out.print(a + " ");
```

```
    c = a + b;
```

```
    a = b;
```

```
    b = c;
```

```
}
```

```
break;
```

```
}
```

case 4:

```
System.out.println("enter case 1-4");
```

```
int dchoice = sc.nextInt();
```

```
switch (dchoice) {
```

case 1:

```
System.out.print("Enter a number: ");
```

```
int n = sc.nextInt();
```

```
int fact = 1;
```

```
for (int i = 1; i <= n; i++) {
```

```
    fact = fact * i;
```

```
}
```

```
System.out.println("Factorial = " + fact);
```

```
break;
```

case 2:

```
System.out.print("Enter a number: ");
```

```
int num = sc.nextInt();
```

```
int count = 0;
for (int i = 1; i <= num; i++) {
    if (num % i == 0) {
        count++;
    }
}

if (count == 2)
    System.out.println("Prime Number");
else
    System.out.println("Not a Prime Number");
break;
```

case 3:

```
System.out.print("Enter first number: ");
int a = sc.nextInt();

System.out.print("Enter second number: ");
int b = sc.nextInt();

if (a > b)
    System.out.println("Maximum = " + a);
else
    System.out.println("Maximum = " + b);
break;
```

case 4:

```
System.out.print("Enter radius: ");
```

```
double r = sc.nextDouble();
```

```
double area = 3.14 * r * r;
```

```
System.out.println("Area of Circle = " + area);
```

```
break;
```

default:

```
System.out.println("Invalid Choice");
```

```
}
```

```
break;
```

```
}
```

```
}
```

```
}
```

OUTPUT SCREENSHOT (SAMPLE RUN):

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 1

Enter case 1-4: 4

Enter Student Name: Sameera

Enter Roll Number: 102

Enter English Marks: 85

Enter Maths Marks: 90

Enter Social Marks: 80

----- RESULT -----

Name : Sameera

Roll No : 102

Total : 255.0

Percentage : 85.0